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Batch: A1
Division: A

Subject: Programming with Java

ASSIGNMENT No : 5

TITLE : Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Output Screenshots

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c:; cd 'C:\Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode\ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redhat'
Fruits : [Apple, Grapes, Mango]
New Fruits : [Apple, Grapes, Mango]
Size : 3
Contains Apple? : true
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5>
Element : 20
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c:; cd 'C:\Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode\ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redhat'
Vector : [10, 30, 40]
New Vector : [30, 40]
Size : 2
Element : 30
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5>
```

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> & 'c:\
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Nilima\Ap
edhat.java\jdt_ws\LAB EXP 5_8277c32d\bin' 'Main3'
```

- Original String : Programming
- After Append : Programming in Java
- After Insert : Programming Languagein Java
- After Replace : Coding Languagein Java
- After Delete : Codingein Java
- After Reverse : avaJ niegnidoC

```
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5>
```

JavaScript

```
● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c:; cd 'c
Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowC
ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redha
```

Tokens are :

R

C++

Python

SQL

Java

```
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5>
```

Exercise

Q1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c::; cd 'c:\Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode\ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redhat
----- TO DO List -----
1.Nap
2.Practice problems
3.Study
4.Build a project
5.Coding
6.Review notes
7.Play a sport
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5>
```

Q2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c::; cd 'c:\Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode\ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redhat
Registered Courses
-----
1. Computer Science
2. Artificial Intelligence
3. Data Science
4. Cyber Security
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5>
```

Conclusion

This assignment helped in understanding the use of ArrayList, Vector, and StringBuffer in Java for dynamic data storage, management, and string manipulation. It demonstrated how ArrayList and Vector can efficiently store and manage collections of data, while StringBuffer enables efficient modification of strings without creating new objects. The implementation improved my understanding of the Java Collections Framework, dynamic memory management, and mutable string operations. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, database management systems, and other real-world Java applications that require efficient data handling and string processing.